Igneous Rocks



Igneous rocks are formed when molten rock (magma) cools and solidifies either below (intrusive) or above (extrusive) the Earth's surface. The rate at which the magma cools determines the rock's properties. An igneous rock can take minutes or millions of years to form, depending on its rate of solidification and cooling of magma. Magma itself takes millions of years to be created as tectonic plates slide under each other and melt.

Intrusive igneous rocks such as diorite, gabbro and granite have coarse grains. They have larger crystals than extrusive rocks because of a slower cooling and solidification rate below the surface.

Extrusive igneous rocks such as basalt, obsidian and pumice have much smaller crystal formations than intrusive igneous rocks because of the faster cooling rates that occur on the surface.

Give the meaning of the following words in the paragraphs above:

1. Molten _____

2. Solidifies _____

3. Intrusive _____

4. Extrusive _____

5. Properties _____

6. Magma _____

7. Coarse _____

Which words or phrases can be replaced with the following:

1. Made _____

2. Goes hard _____

3. Speed _____

4. Characteristics _____

5. Produced _____

6. Rough _____

7. Happen _____



Questions

1. How are igneous rocks formed? _____

_____

2. What determines a rock's properties? _____

_____

3. How long does an igneous rock take to form? _____

_____

4. How long does magma take to form? _____

_____

5. Name two intrusive igneous rocks _____

6. How are intrusive igneous rocks different to extrusive igneous rocks? _____

_____

_____

7. Name two extrusive igneous rocks _____

8. Why do extrusive igneous rocks have small crystals? _____

_____

